Module 02

DevOps Roadmap

DevOps end to end services & solutions

Agenda

- What is DevOps?
- Why do we need DevOps?
- DevOps Lifecycle
- DevOps Technology Stack
- DevOps Tools
- DevOps Culture
- Case Studies: Netflix, Amazon, Etsy

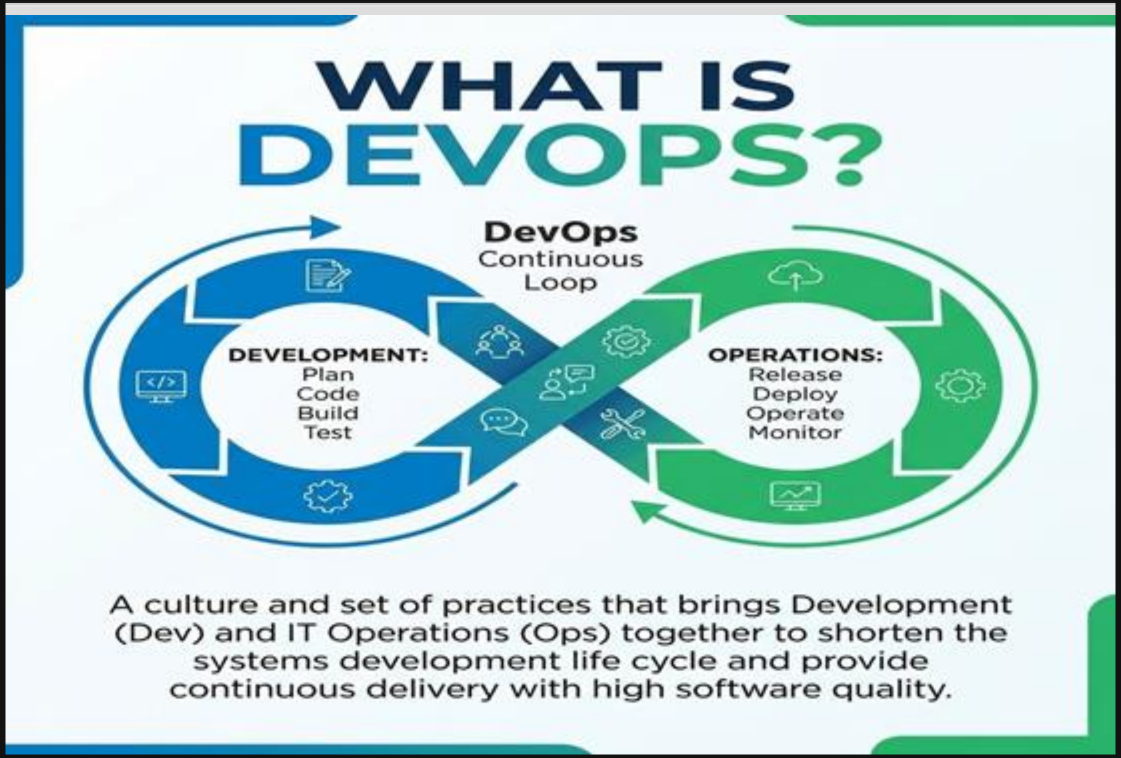
What is DevOps?

Software Development Life Cycle(SDLC)

Definition

A set of practices that combines software development (Dev) and IT operations (Ops).

- **Goal:** Shorten the development lifecycle & deliver high-quality software continuously.
- **Transformation:** It creates cultural changes by transforming how operations, developers, and testers collaborate.



Why do we need DevOps?

Speed

Move at high velocity so you can innovate for customers faster.

Reliability

Ensure the quality of application updates and infrastructure changes.

Scale

Operate and manage your infrastructure and development processes at scale.

Collaboration

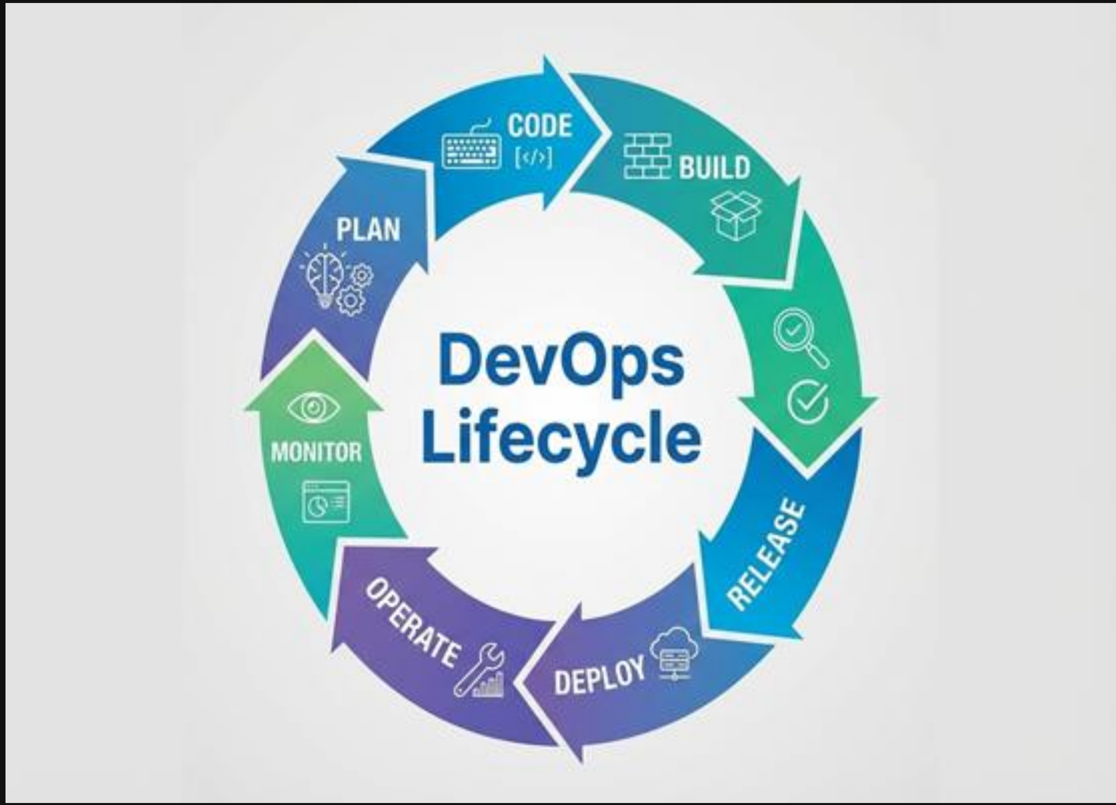
Build more effective teams under a DevOps cultural model.

DevOps Lifecycle

∞Continuous Loop

The lifecycle consists of phases representing the continuous nature of DevOps.

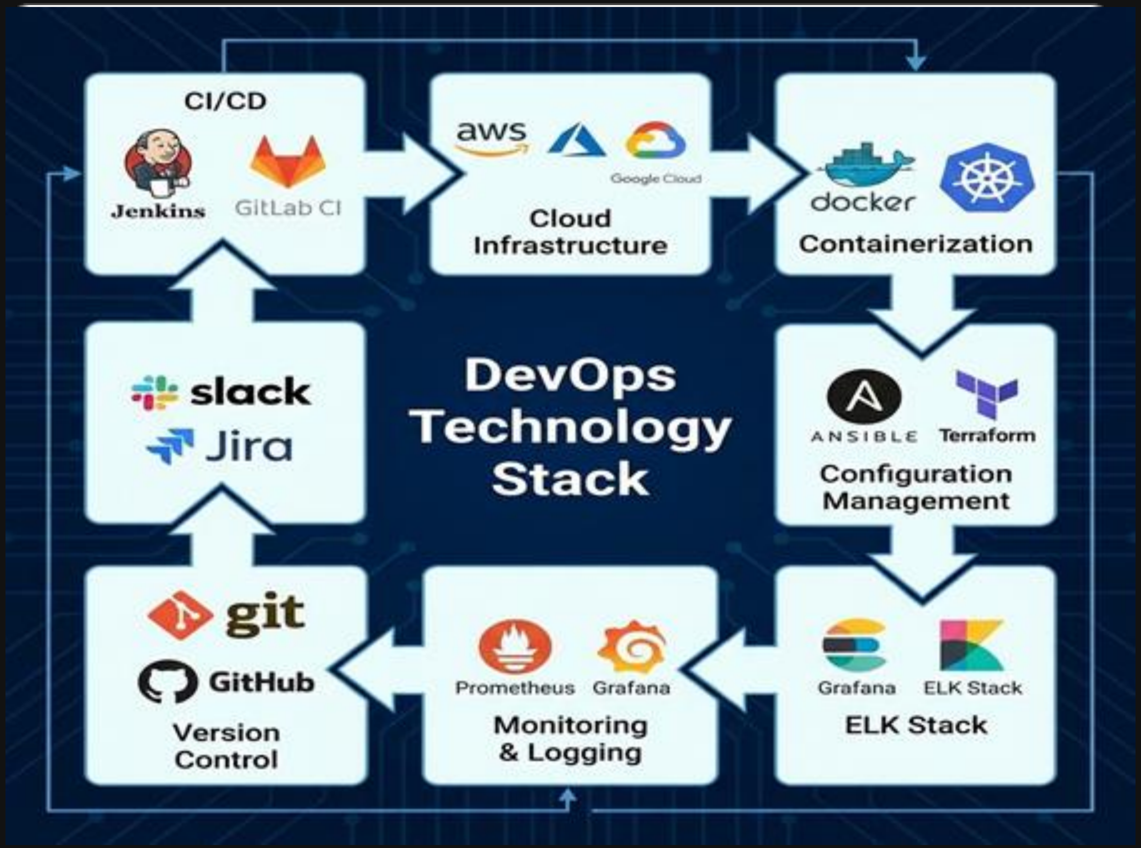
- **Plan & Code:** Development phase.
- **Build & Test:** CI (Continuous Integration).
- **Release & Deploy:** CD (Continuous Delivery).
- **Operate & Monitor:** Feedback loop.



DevOps Technology Stack

Essential Tooling

- **Planning:** Jira, Trello
- **SCM:** Git, GitHub, GitLab
- **CI/CD:** Jenkins, CircleCI, GitLab CI
- **Containerization:** Docker, Kubernetes
- **IaC:** Terraform, Ansible
- **Monitoring:** Prometheus, Grafana



DevOps Culture

Shared Responsibility

Developers and Operations share responsibility for the product.

Automation

Automate repetitive tasks to focus on innovation.

Measurement

Measure everything to understand and improve the process.

Success Stories

🎬Netflix

Pioneers of "Chaos Engineering" (Chaos Monkey). They deploy code thousands of times daily using Spinnaker.

a Amazon

Moved to microservices early. Famous for their "Two Pizza Teams" rule and fully automated deployments.

E Etsy

Fully embraced continuous deployment, deploying 50+ times a day with a strong focus on monitoring.

Implementation

Challenges

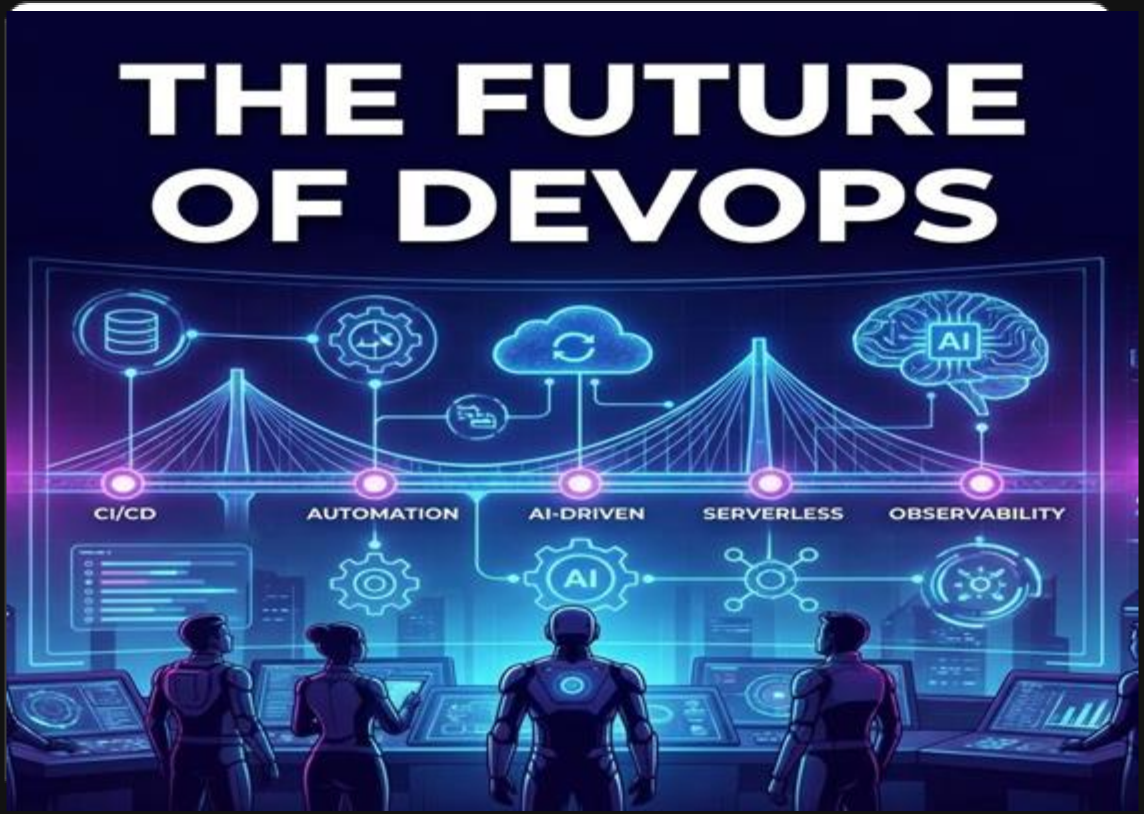
- Cultural Resistance to change.
- Skill Gaps in teams.
- Tool Overload/Complexity.

Getting Started

- **Assess:** Evaluate current state.
- **Start Small:** Pick a pilot project.
- **Automate First:** Identify easy wins.
- **Collaborate:** Break down silos.

The Future of DevOps

- **AI/ML in DevOps:** Using AI to automate and optimize pipelines (AIOps).
- **DevSecOps:** Security integrated from the start.
- **Serverless & Microservices:** Architectures offering greater scalability.
- **GitOps:** Using Git as the single source of truth for infrastructure.



Future of DevOps

AI/ML

AIOps for predictive analytics and automation.

Serverless

Moving towards serverless architectures.

GitOps

Managing infrastructure through Git repositories.

Homework - **see the Podcast**

Getting Started

- https://www.youtube.com/watch?v=SCZQFmFDugQ

Questions?

Thank you for participating.

Feel free to reach out:
https://dinghy.co.il